

Scarlett Enos

CS 499

05/30/2026

Data Structures and Algorithms

This artifact is from a personal project I did to complete a certification, and it consists of a database project for a clothing store inventory system. The original project focused on creating a database table, inserting records, and writing basic SQL queries to retrieve and sort inventory information. I originally created this artifact to decide if I wanted to pursue data analytics as a career back at the beginning of my academic journey with SNHU.

The reason I chose this artifact is because I believe it demonstrates my growth in working with databases, organizing data, and using SQL to solve problems. The original artifact was very simplistic and showed basic skills such as creating tables, inserting records, and filtering and sorting data. These skills are important when working with algorithms and data structures because they involve organizing information and retrieving information. To improve the artifact, I expanded the database structure by adding categories and reorder level fields, I added additional inventory records, and created new queries that calculate inventory value, identify items that are low in stock, group products by category and fabric type, and prioritize low stock items in need of being reordered. I also improved my documentation by adding more inline comments throughout my code.

I believe that I met the course outcomes because my enhancements improved the way data is organized and analyzed. The updated artifact demonstrates my ability to work with larger datasets and use SQL to retrieve information from the data. Currently, I do not plan to make any

more significant changes to this artifact because I believe my enhancements successfully show my skills in algorithms and data structures.

While making enhancements, I learned that making small changes to a database can make the information much more useful. Adding fields like category and reorder level helped me create better queries for sorting, grouping, calculating, and retrieving data. One challenge I faced was working through errors in SQLiteStudio, but fixing those issues helped me better understand how to use the platform to create databases.